EDGE Artix 7 Board File Installation

Vivado board files edgeA7, edgeA7_50t and edgeA7_100t · Vivado 2015.1 or later · Windows and Linux

A board file tells Vivado which FPGA is fitted to the board and how the clock, switches, LEDs, UART and other peripherals are connected. When the board files are installed, you can select the EDGE board instead of an FPGA part when you create a project, and the Board tab in IP Integrator can connect peripherals for you, as in the MicroBlaze guide.

Note: HDL projects that use the EDGE Artix 7 master XDC file do not need a board file. Selecting the board still sets the correct FPGA part for you.

Which board file do I need?

The three EDGE Artix 7 variants use the same board with a different FPGA, so each has its own board file. The part number is printed on top of the FPGA, and Hardware Manager shows the device name when the board is connected.

Board variant	FPGA	Board file folder	Vivado part
EDGE Artix 7 35T	XC7A35T-1FTG256C	`edgeA7`	`xc7a35tftg256-1`
EDGE Artix 7 50T	XC7A50T-1FTG256C	`edgeA7_50t`	`xc7a50tftg256-1`
EDGE Artix 7 100T	XC7A100T-1FTG256C	`edgeA7_100t`	`xc7a100tftg256-1`

Each folder contains a version folder (1.0) with board.xml, part0_pins.xml and preset.xml. Keep this structure when you copy them.

Interfaces in the Board tab (edgeA7_50t and edgeA7_100t)

Board tab entry	Connected to	IP it creates
System Clock	50 MHz oscillator, pin N11	Clocking Wizard
CPU Reset	Centre push button SW22, pin M14, active high	Processor System Reset
16 Slide Switches	SW0 to SW15, same bit order as the master XDC	AXI GPIO
16 LEDs	D3 to D19 except D7 (Bluetooth status LED)	AXI GPIO
4 Push Buttons	Up, Left, Right and Down buttons	AXI GPIO
7 Segment Display	4 digit enables and 8 segments (shared with SDRAM address lines)	AXI GPIO
QSPI Flash	64 Mb configuration flash	AXI Quad SPI
USB UART	FT2232H USB-UART, 115200 baud	AXI UART Lite
WiFi UART	ESP8266-12F module (fit jumper J3)	AXI UART Lite
Bluetooth UART	HM-10 module (fit jumper J4)	AXI UART Lite

Note: The edgeA7 board file for the 35T board sets the USB UART to 9600 baud. The 50T and 100T board files use 115200 baud.

Get the board files

Download the board files from either place below. Both contain the same files.

Source	Where	How to download
GitHub	github.com/AllAboutFPGA/EDGE-FPGA-Kit-Board-Files	Click Code > Download ZIP, then extract the zip file. It also contains the board files for the other EDGE boards.
Dropbox	EDGE Artix 7 Board Documents > board_files > New_Board_File_Vivado_2015.1_and_above	Open the folder and click Download. Dropbox saves the folder as a zip file; extract it.

You need only the **edgeA7**, **edgeA7_50t** and **edgeA7_100t** folders. Install them with method A or method B below.

Method A: one folder for all Vivado versions (recommended)

This method keeps the board files outside Vivado, so they keep working when you install a new Vivado version.

Step 1: Create a permanent folder, for example `C:\EDGE\board_files`, and copy the **edgeA7**, **edgeA7_50t** and **edgeA7_100t** folders into it.

Step 2: Open the Vivado settings folder. On Windows, type `%APPDATA%\Xilinx\Vivado` in the File Explorer address bar. On Linux, use `~/.Xilinx/Vivado`. Create the folders if they do not exist.

Step 3: In that folder, create a text file named **Vivado_init.tcl** with this line, using forward slashes in the path:

```
set_param board.repoPaths [list "C:/EDGE/board_files"]
```

In Notepad, set Save as type to **All Files** so the file is not saved as Vivado_init.tcl.txt. If you already have a Vivado_init.tcl, for example for other vendors' boards, add the EDGE folder to the same list:

```
set_param board.repoPaths [list "C:/EDGE/board_files"
"C:/other/board_files"]
```

Step 4: Restart Vivado. Vivado runs Vivado_init.tcl every time it starts.

Note: Vivado also reads Vivado_init.tcl from a folder named after the version, for example `%APPDATA%\Xilinx\Vivado\2026.1`. If that file exists, add the line to it as well.

Method B: copy into the Vivado installation

Step 1: Close Vivado. Open the installation folder for your Vivado version (the folder that contains the **bin** and **data** folders, for example `C:\Xilinx\Vivado\2024.2`).

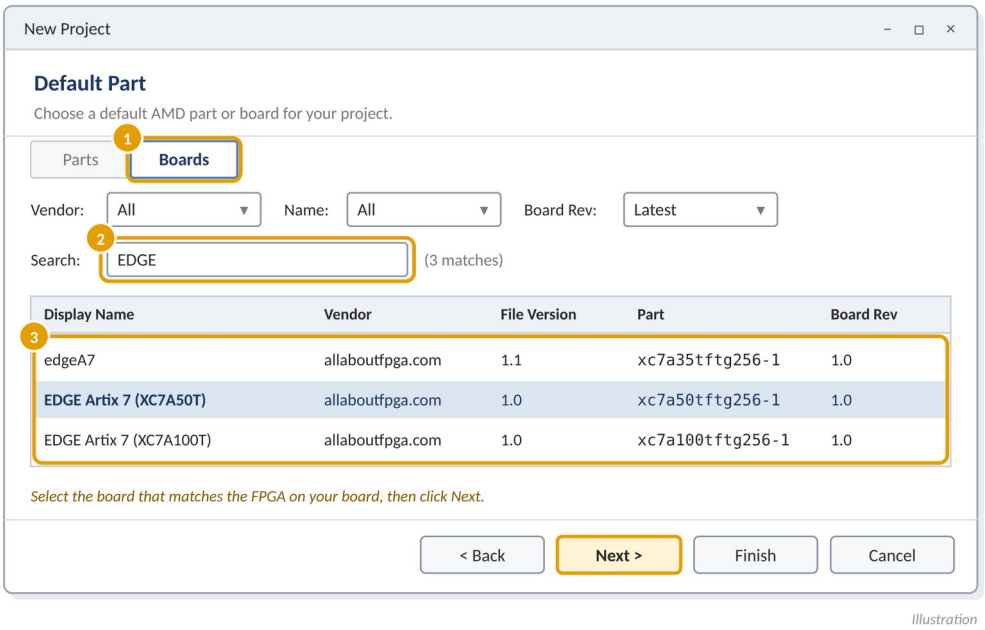
Step 2: Open `data\boards\board_files`. Create the board_files folder if it does not exist.

Step 3: Copy the **edgeA7**, **edgeA7_50t** and **edgeA7_100t** folders into it. On Linux, you may need `sudo` to write to the installation folder.

Step 4: Repeat for each Vivado version you use, then restart Vivado.

Check that the board files are installed

Step 1: In Vivado, click **Create Project** and go to the Default Part page. Click **Boards** and type **EDGE** in the Search box. The EDGE Artix 7 boards are listed with their parts (xc7a35tftg256-1, xc7a50tftg256-1 and xc7a100tftg256-1). The 35T board is listed as edgeA7. Select the board that matches your FPGA.



Illustration

Illustration New Project wizard, Boards tab with the three EDGE Artix 7 boards

Step 2: You can also check in the Tcl Console:

```
get_param board.repoPaths
get_board_parts *edgeA7*
```

The second command lists one board part for each installed EDGE board file.

Troubleshooting

The EDGE boards are not listed. Check the folder structure: board.xml must be in board_files\edgeA7_50t\1.0\, not in an extra folder level created when you extracted the zip file. Check that the file is named Vivado_init.tcl, not Vivado_init.tcl.txt, and that the path in it uses forward slashes.

The bitstream does not program the board. A bitstream built for one variant does not load on another. Compare the device shown in Hardware Manager (xc7a35t_0, xc7a50t_0 or xc7a100t_0) with the project part.

Changing an existing project to another variant. Select Tools > Settings > General, change the Project device to the other EDGE board, and then regenerate the output products and bitstream.